

Day 2

Genetic Load



Outline

1

Inbreeding

The theory of genetic load

Examples

2

Methodology 1: SnpEff

Tutorial – part 1

Methodology 2: GERP

Tutorial – part 2

3

Discussion



The «ingredients» for genetic load

Mutations (and inbreeding)

Mutations

Genetic variants are introduced by:



Mutation



Gene flow



Recombination

Their fate can be determined by two forces that can operate simultaneously:



Genetic drift



Selection

Mutation properties

Mutation effect

$$(-1 \leq s \leq 1)$$

It depends on the consequences on fitness:

- Neutral → no selection
- (weakly/highly) Beneficial → positive selection
- (weakly/highly) Deleterious → negative selection

Dominance coefficient

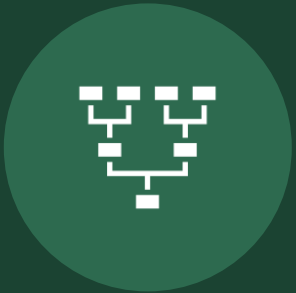
$$(0 \leq h \leq 1)$$

It depends on the phenotypic effect with respect to the genotype:

- Dominant: same effect whether an individual is homozygote or heterozygote
- Recessive: the phenotypic effect manifests only when an individual is homozygote

Inbreeding

It is the mating between relatives



It can result in inbreeding depression:

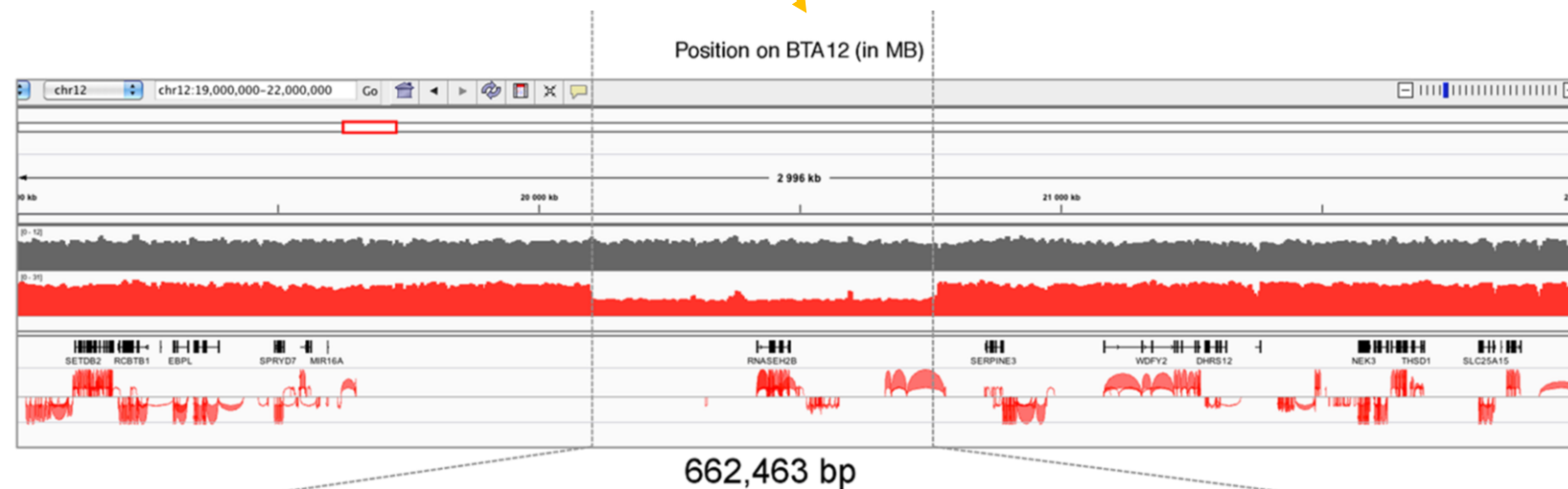
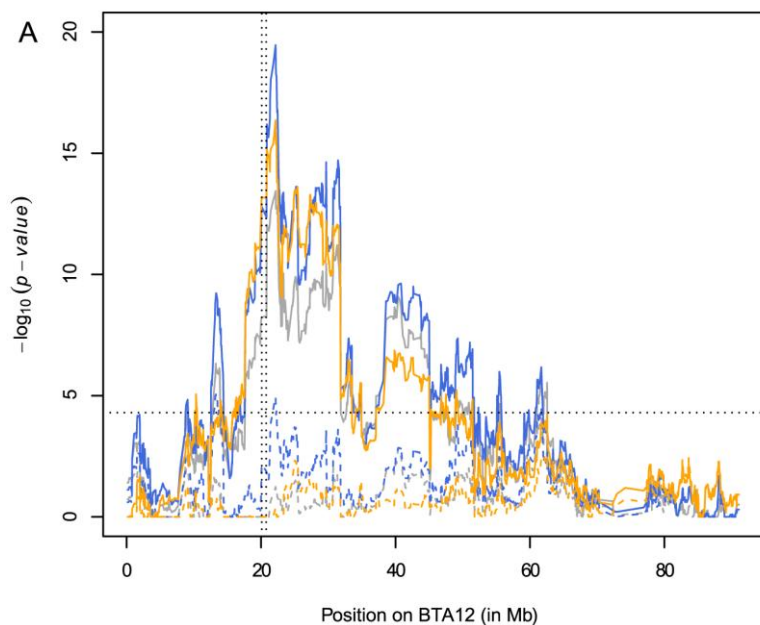
A reduction of fitness that results from mating between relatives, which exposes deleterious recessive alleles

Example 1: breeding cattle



- Artificial selection for breeding exacerbates the mating among relatives
- Highly deleterious variants can come in homozygosity and have a strong effect on fitness

QTL associated with milk production on chromosome 12 at around 20Mb for (direct or indirect) effect from deletion involving 4 genes



Example 2: wild case studies in brief



Florida panther

Heart defects, kinked tails, undescended testes and poor sperm quality



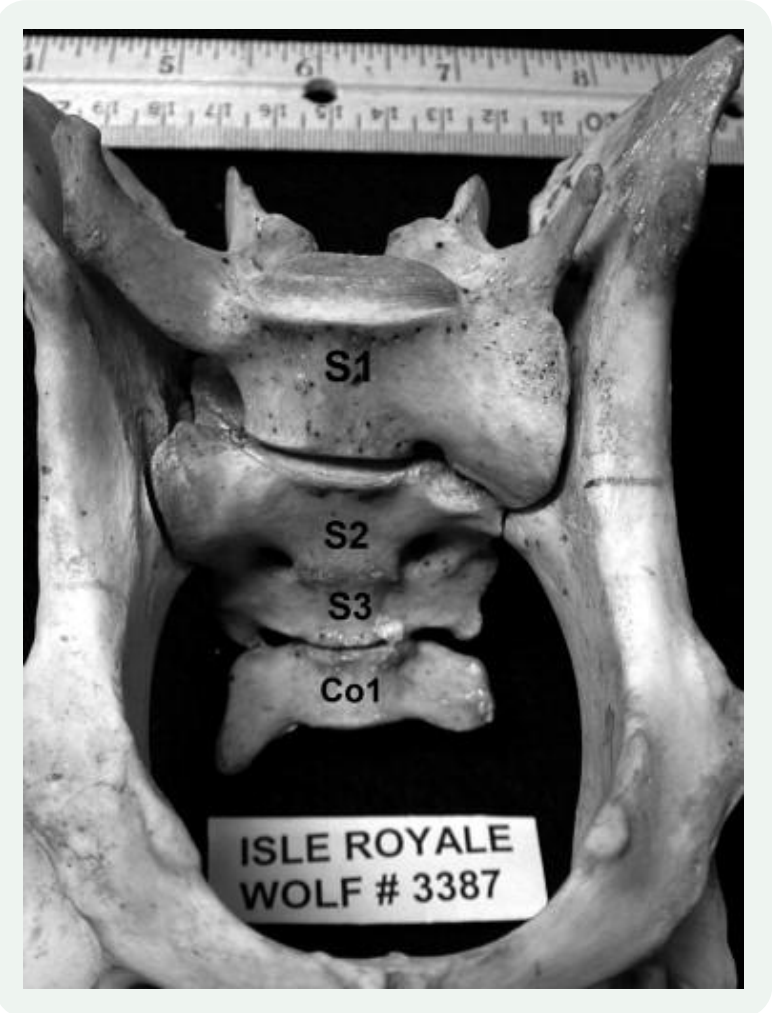
Isle Royale wolves

Bone deformities



Eurasian lynx

Heart defects in the form of oversized hearts



What is the genetic load?

1950

First conceptualization in 1950 by H. J. Muller

1957

The cost of natural selection (Haldane, 1957)

1958

The proportion by which the population fitness is decreased in comparison with an optimum genotype (Crow 1958)

Today

Several definitions with different flavors (e.g. expansion load, inbreeding load, hybrid load)

The components of genetic load

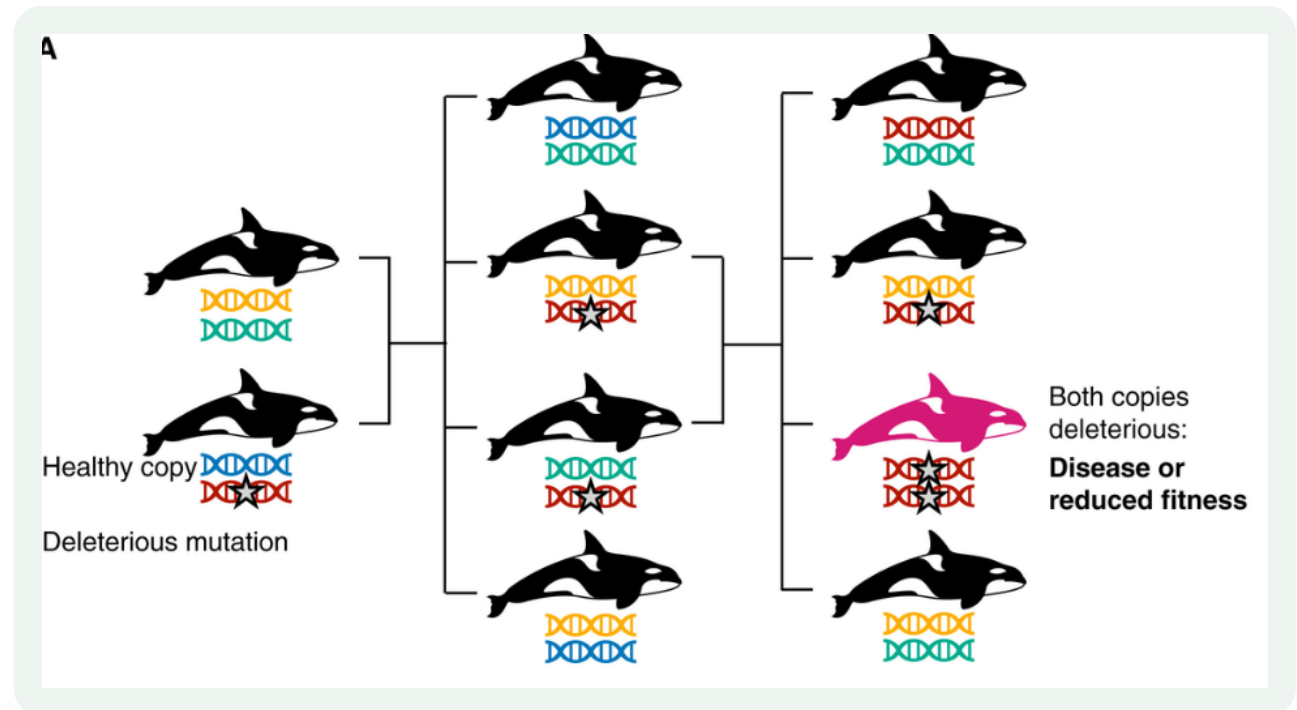
Realized load

It reduces the fitness in the current generation

Masked load

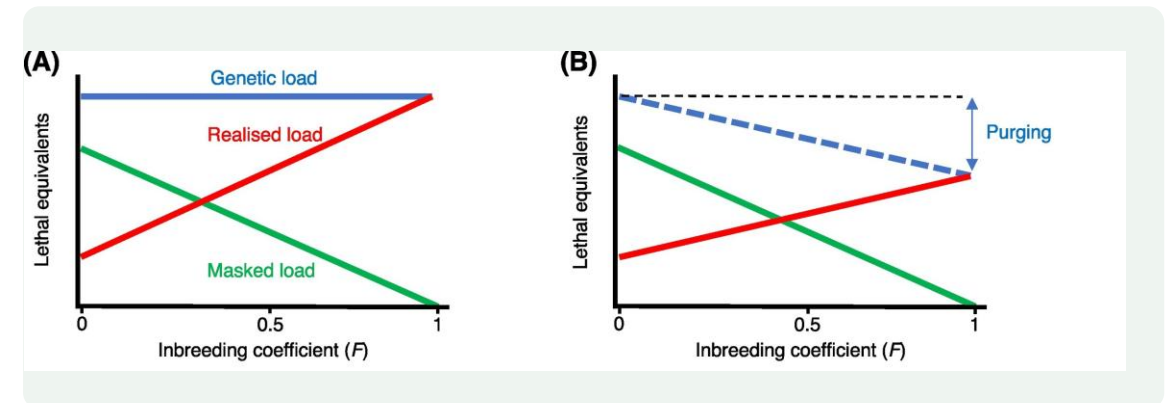
It can reduce fitness in the future

Link with inbreeding →

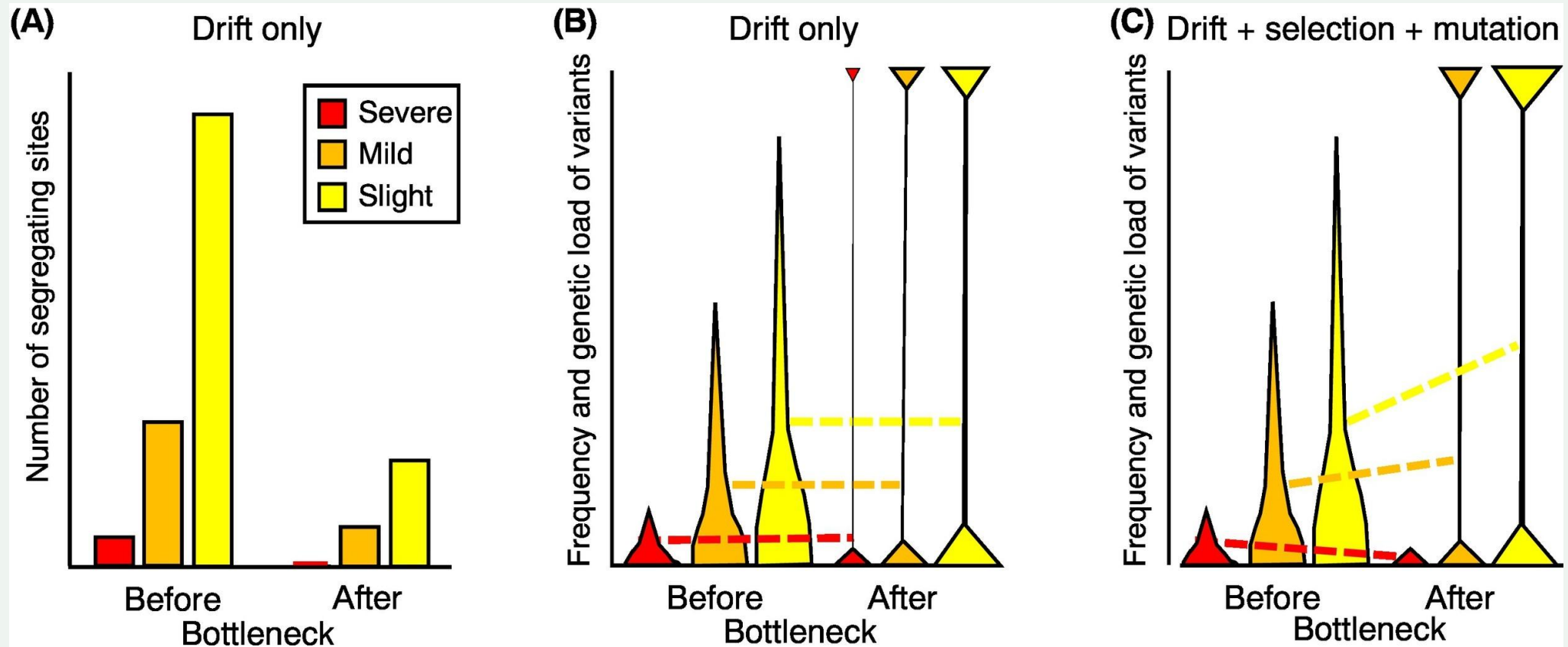


Genetic load and Ne

- Small populations = low Ne, high inbreeding, high homozygosity
 - high realized load
 - low masked load
- Large populations = high Ne, low inbreeding, high heterozygosity
 - low realized load
 - high masked load
- **BUT! Small populations can have an advantage: purging from purifying selection**

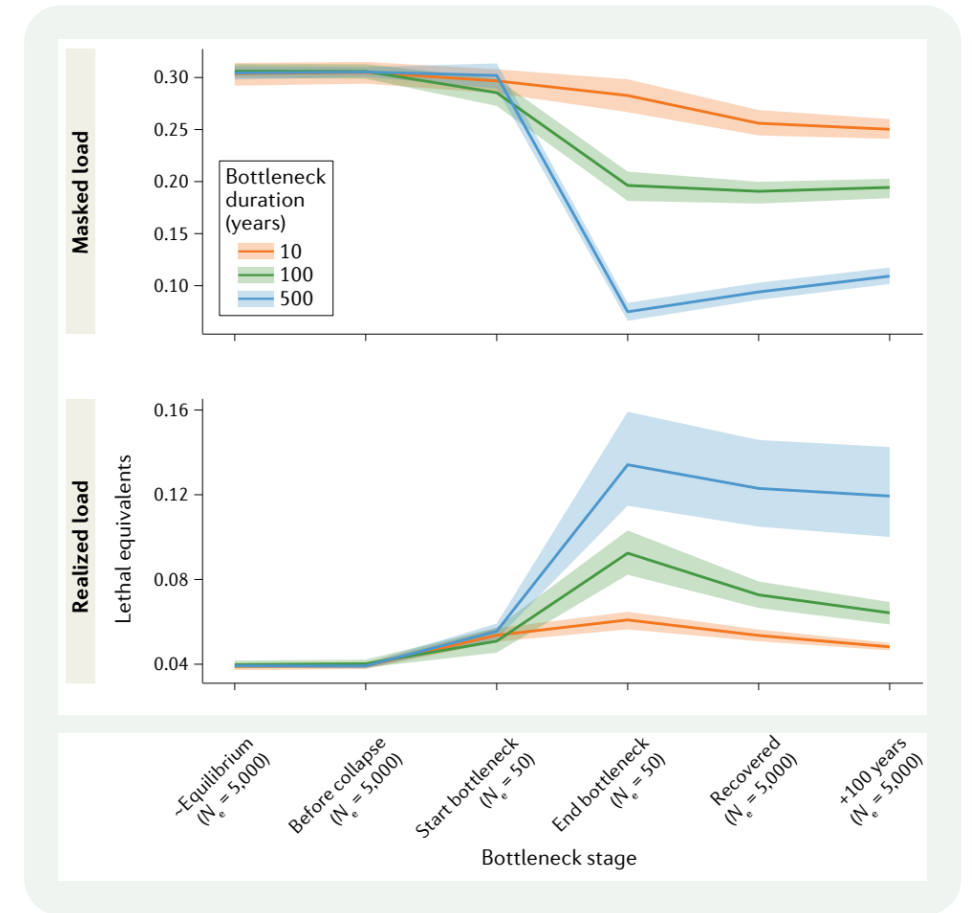


Genetic load and N_e



Genetic load and N_e

- **Bottleneck and population recovery**
- **During bottleneck:**
 - more homozygosity = more realized load
- **After bottleneck:**
 - Increase in N_e = less realized load



How to estimate genetic load?



Focus on fitness

A proportional decrease in the average fitness of a population relative to that of the optimal (maximal) genotype.



Use of genomics without fitness data

Summing up the selection coefficients of all deleterious mutations

Empirical estimate of genetic load

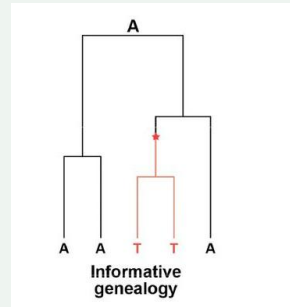
Three steps:

1

Predict deleterious effect of mutations

2

Polarize (i.e. determine the ancestral and derived alleles)



3

Summarize in score(s) and plots

1. Predict deleteriousness



SnpEff

Expected effects based on the type of substitution and its known effects in model species



GERP

Evolutionary conservation across a multispecies alignment

1. Predict deleteriousness - SnpEff

- It is based on biochemical studies (e.g. chemical properties of amino acids) and functional annotations (e.g. substitutions in coding regions)
- Score: low, moderate or high impact mutations
- Limit: genome annotation must be available

```
Scaffold_4      40861672      .      C      T      1525.11 PASS      AC=2;AF=0.022;AN=92;DP=
764;ExcessHet=0.0239;FS=0;InbreedingCoeff=0.9436;MLEAC=2;MLEAF=0.022;MQ=60;QD=27.37;SOR=0.984;A
NN=T|stop_gained|HIGH|UrArMar_g16933|UrArMar_g16933|transcript|UrArMar_g16933.t1|protein_coding
|6/8|c.514C>T|p.Arg172*|514/654|514/654|172/217||;LOF=(UrArMar_g16933|UrArMar_g16933|1|1,00);NM
D=(UrArMar_g16933|UrArMar_g16933|1|1,00)      GT:AD:DP:GQ:PL
```

1. Predict deleteriousness - SnpEff

- It is based on biochemical scores and functional annotations (e.g. amino acids) and
- Score: low, moderate or high
- Limit: genome annotation not complete

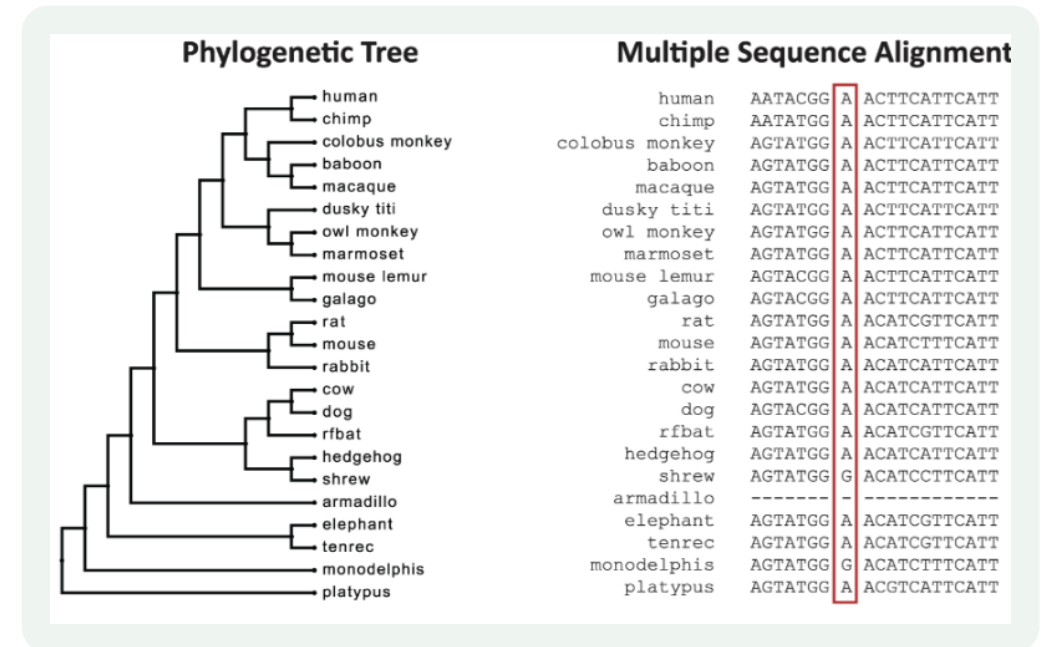
TTT Phe	TCT Ser	TAT Tyr	TGT Cys
TTC Phe	TCC Ser	TAC Tyr	TGC Cys
TTA Leu	TCA Ser	TAA stop	TGA stop
TTG Leu	TCG Ser	TAG stop	TGG Trp
CTT Leu	CCT Pro	CAT His	CGT Arg
CTC Leu	CCC Pro	CAC His	CGC Arg
CTA Leu	CCA Pro	CAA Gln	CGA Arg
CTG Leu	CCG Pro	CAG Gln	CGG Arg
ATT Ile	ACT Thr	AAT Asn	AGT Ser
ATC Ile	ACC Thr	AAC Asn	AGC Ser
ATA Ile	ACA Thr	AAA Lys	AGA Arg
ATG Met	ACG Thr	AAG Lys	AGG Arg
GTT Val	GCT Ala	GAT Asp	GGT Gly
GTC Val	GCC Ala	GAC Asp	GGC Gly
GTA Val	GCA Ala	GAA Glu	GGA Gly
GTG Val	GCG Ala	GAG Glu	GGG Gly

```
Scaffold_4      40861672
764;ExcessHet=0.0239;FS=0;Inbre
NN=T|stop_gained|HIGH|UrArMar_g
|6/8|c.514C>T|p.Arg172*|514/654
D=(UrArMar_g16933|UrArMar_g16933|
```

```
AC=2;AF=0.022;AN=92;DP=
=60;QD=27.37;SOR=0.984;A
g16933.t1|protein_coding
rArMar_g16933|1|1,00);NM
```

1. Predict deleteriousness - GERP

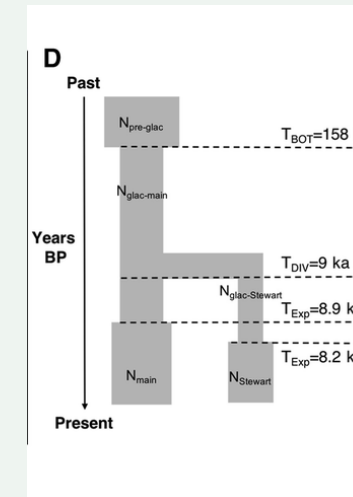
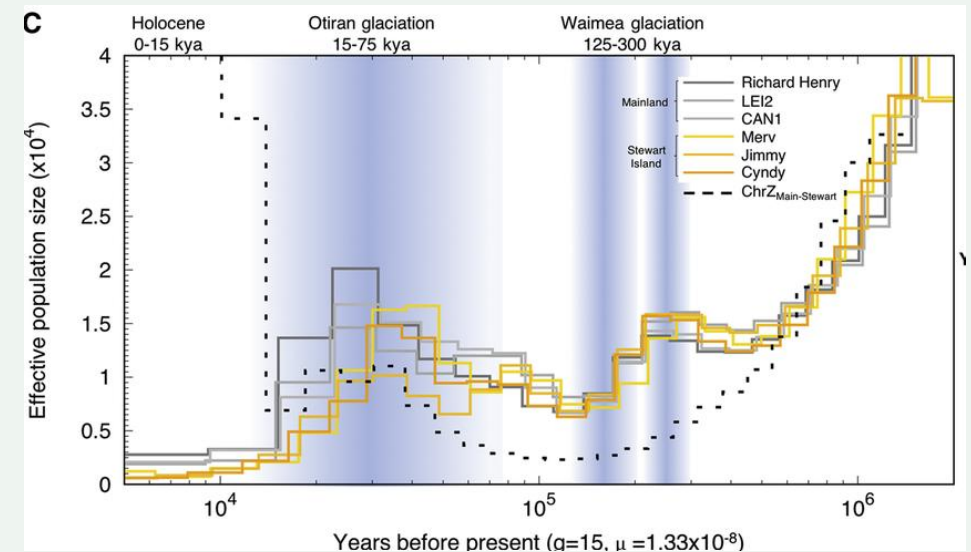
- Genomic Evolutionary Rate Profiling
- Mainly focused on point mutations but indel and SV can be considered too
- Score: the level of conservation in the alignment reflects its functional importance
- Accuracy increases with more species and a phylogenetically broad dataset
- Pros: no annotation
- Cons: turnover of sites, changes in selection pressures, computationally intensive



Example 3: conservation program for the kākāpō

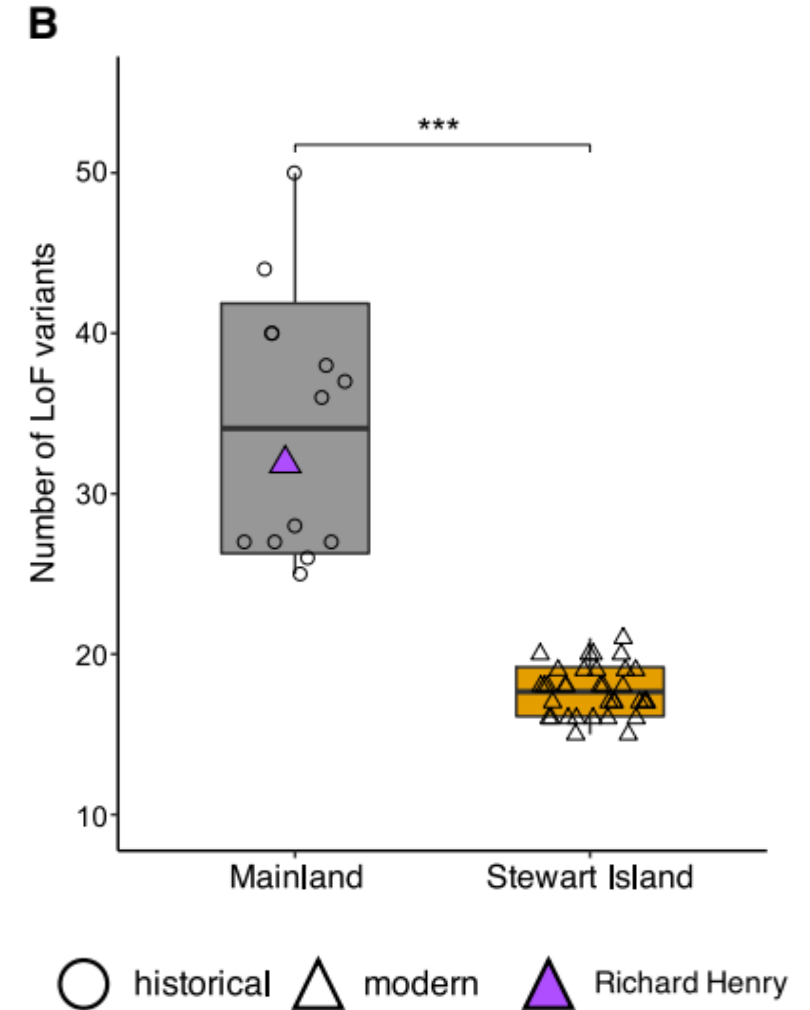
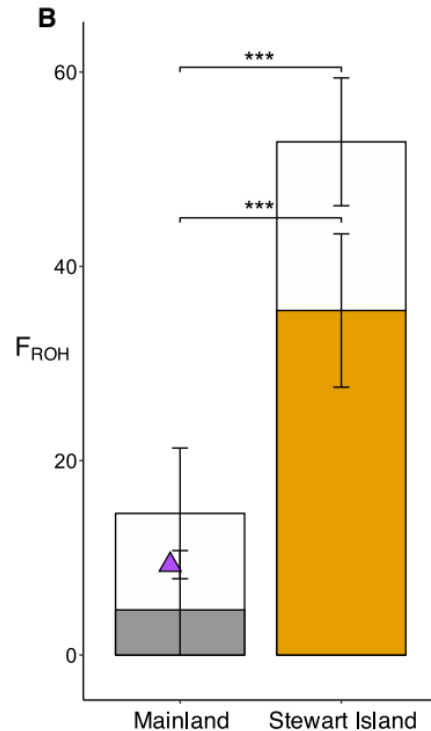
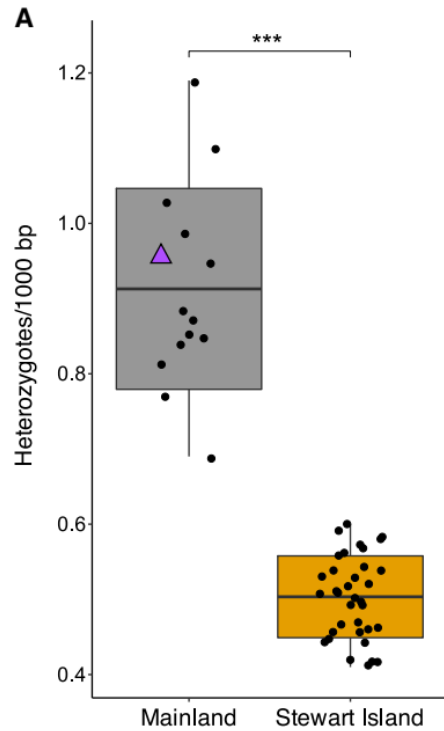
Do you remember from yesterday?

- New Zealand colonizations: 1360 CE and 1800s
- Overhunting and introduction of mammalian predators
- By 1995, the species was reduced to 51 birds
- In 2021, a total of 201 kākāpō in island sanctuaries
- Drop in population well before human settlements
- Stewart Island was colonized much earlier than previously thought

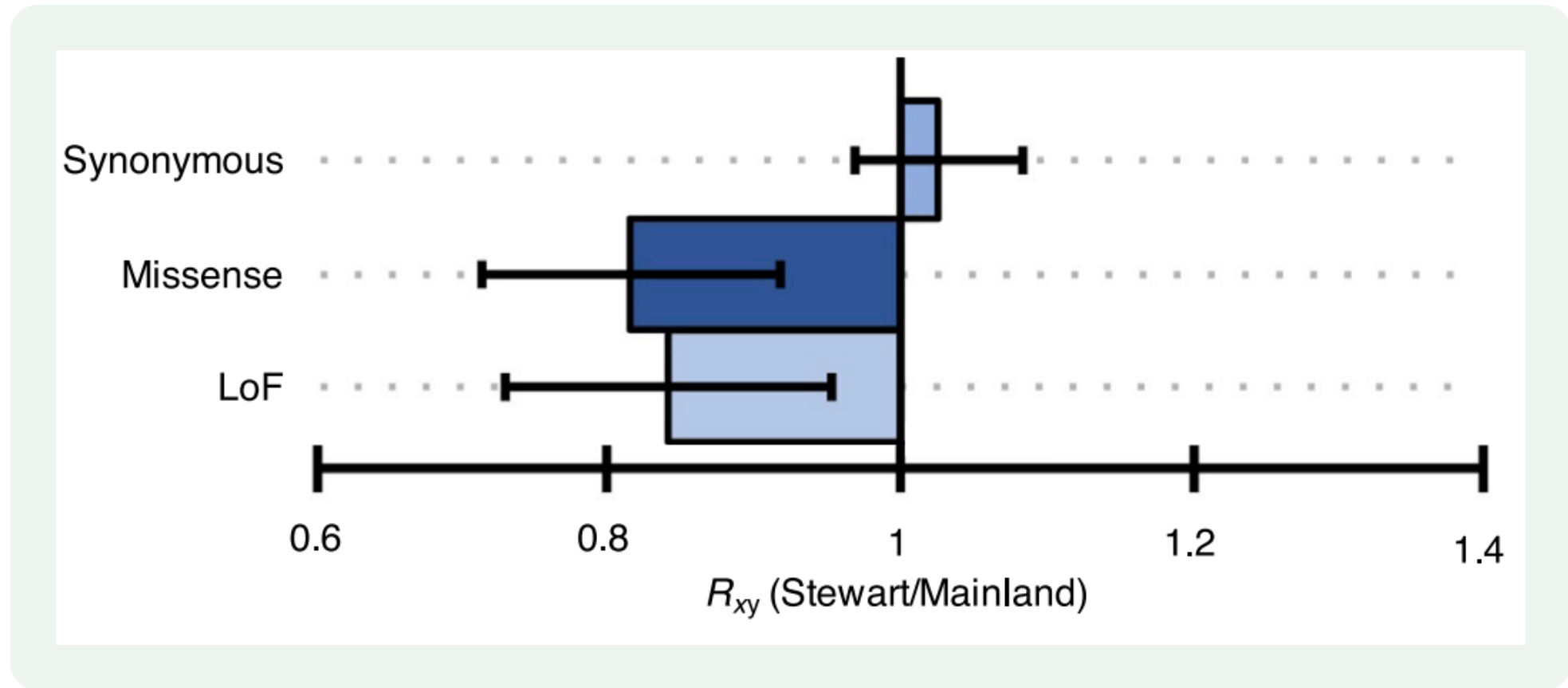


Example 3: conservation program for the kākāpō

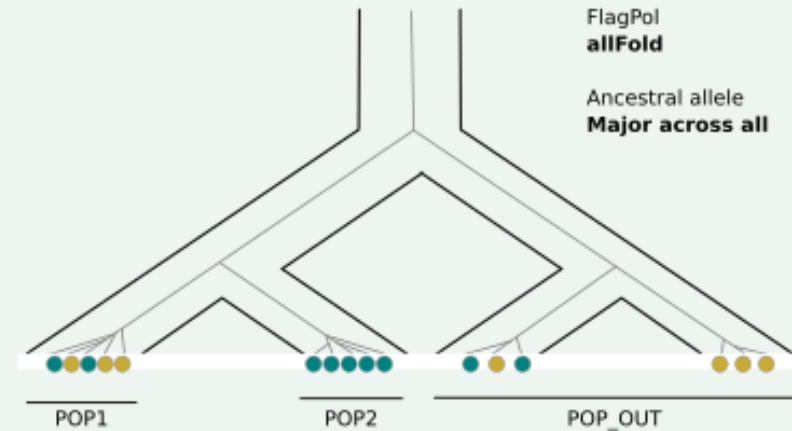
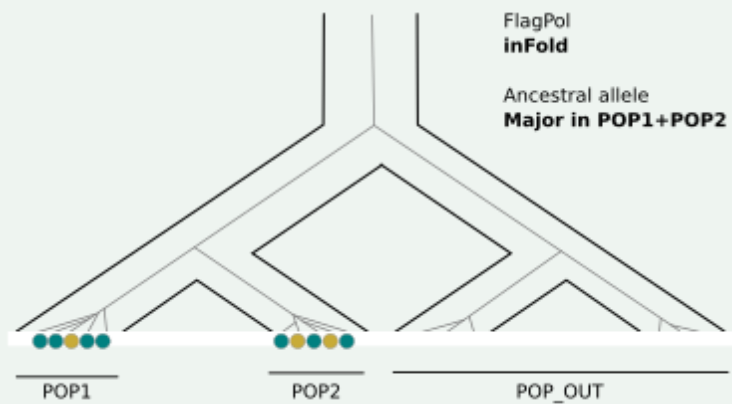
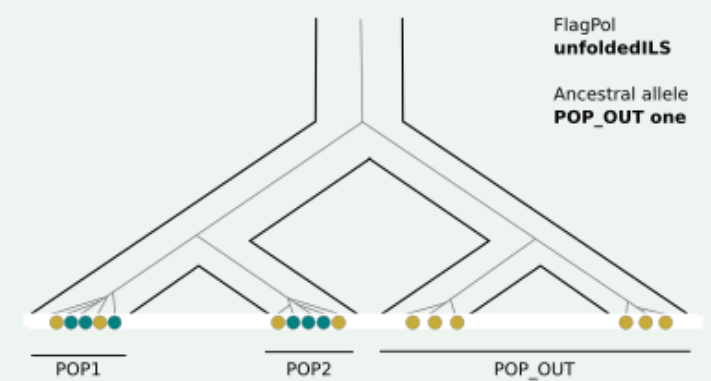
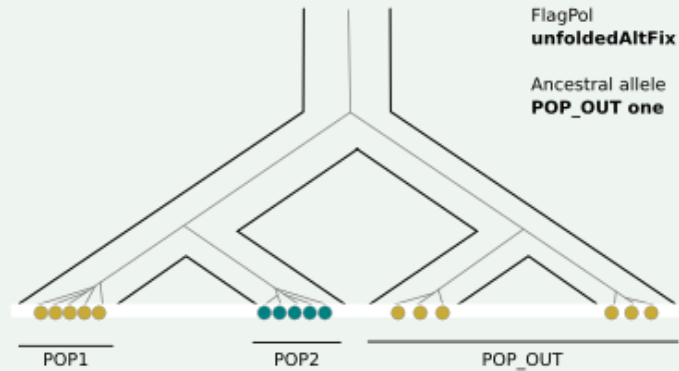
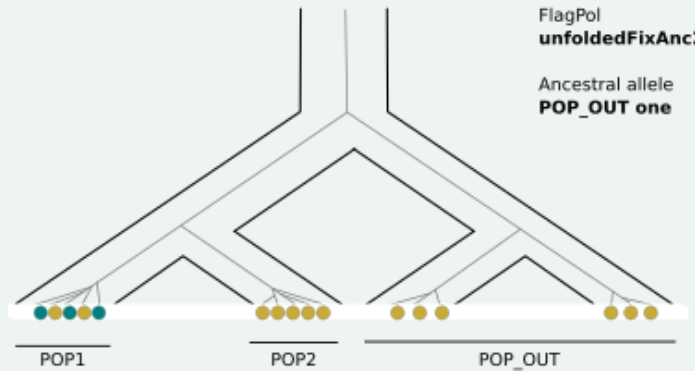
Inbreeding & Genetic load



Example 3: conservation program for the kākāpō



2. Polarization



3. Genetic load proxies or scores

Proxies of the genetic load based on the deleteriousness scores

(Most of the indices below can be separately computed at homozygous or heterozygous sites, within different classes of allelic frequencies and within or outside runs of homozygosity)

Number or proportions of:

- non-synonymous sites
- derived alleles

Ratio between:

- variation at non-synonymous (or zerofold degenerate) sites and variation at synonymous (or fourfold degenerate) sites
- number of private mutations in different populations

Number or proportions of:

- fixed deleterious alleles
- segregating deleterious alleles

Ratio between:

- number of high-damage sites and number of low-damage sites
- number of sites with deleterious mutations and number of synonymous variants
- number of private deleterious mutations in different populations
- number of deleterious alleles in homozygotes versus heterozygotes genotypes

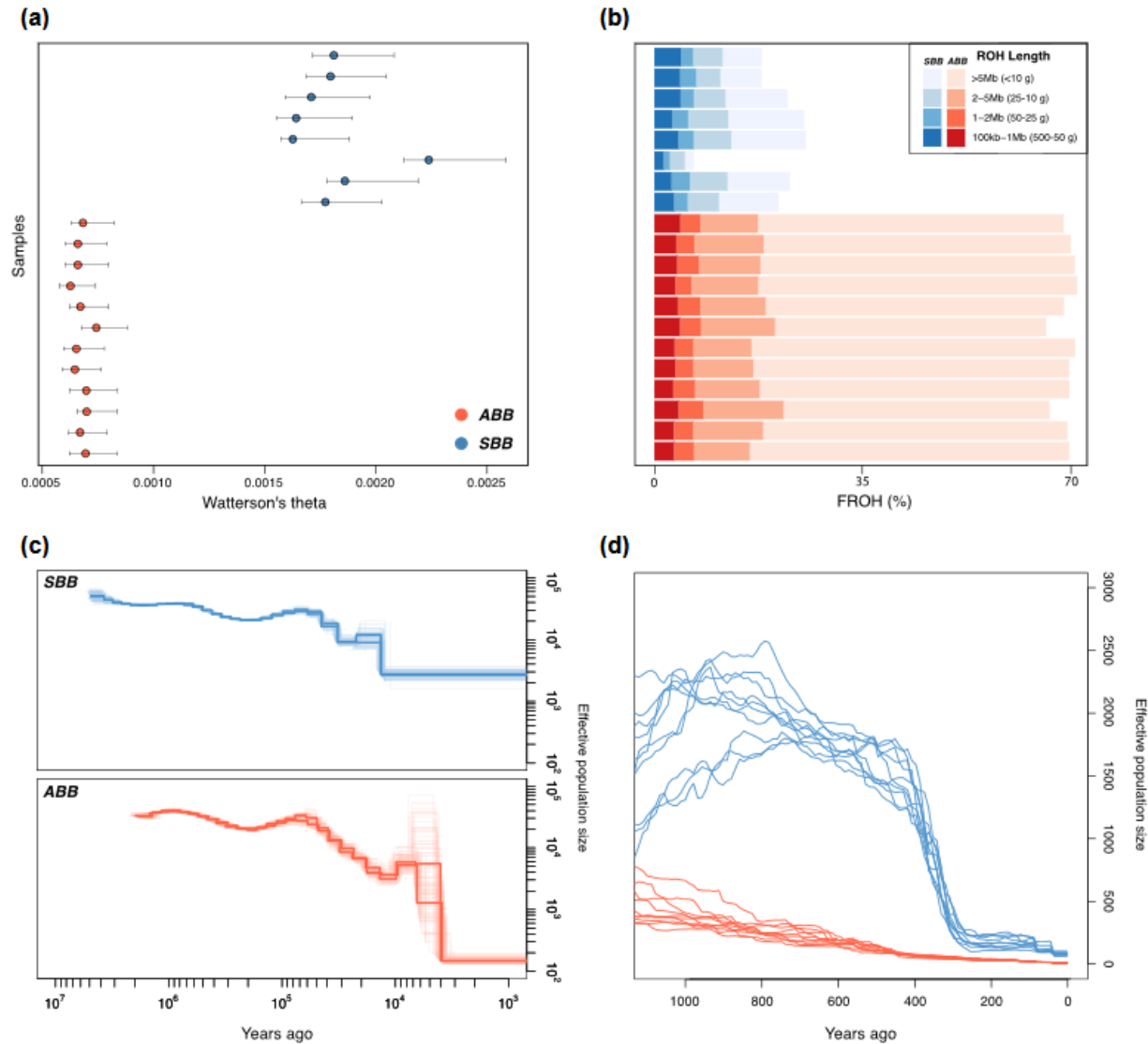
SFS only at deleterious alleles

Sum of scores

Averages across:

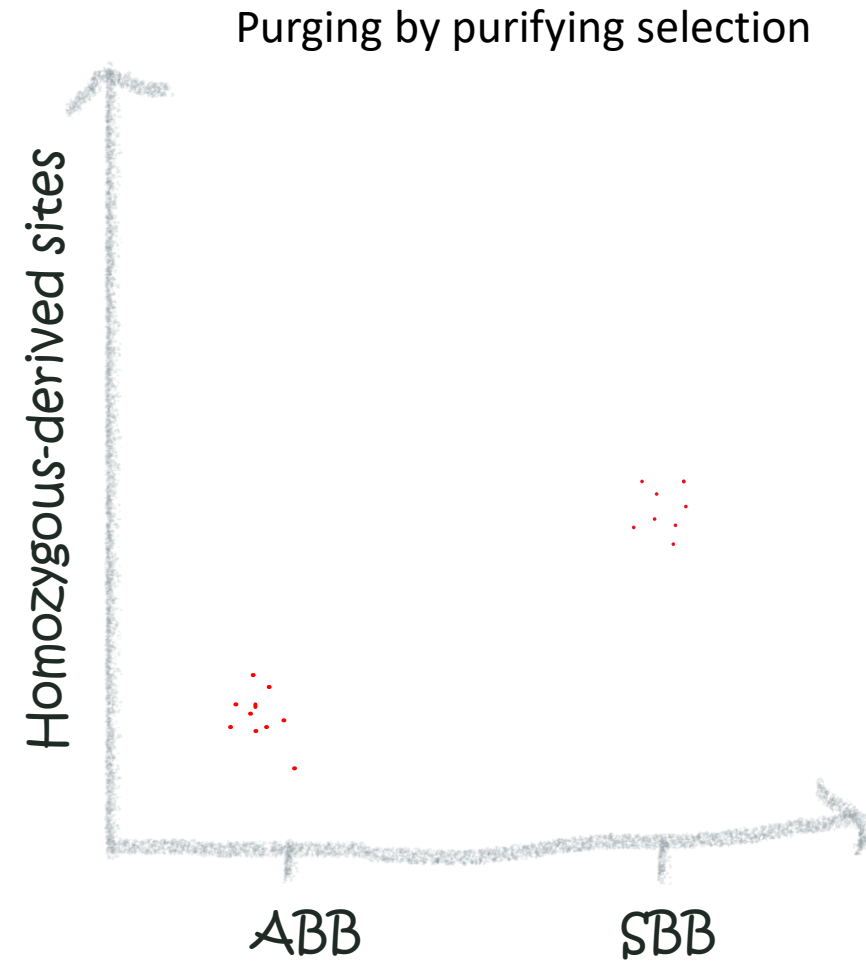
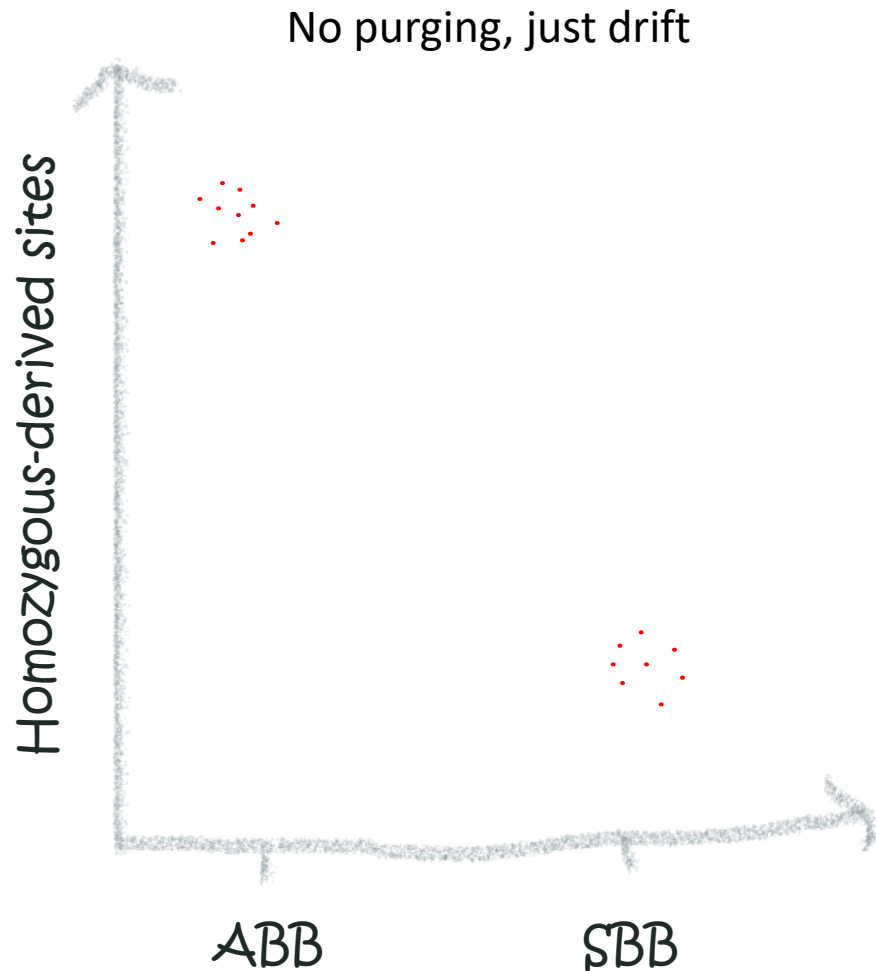
- all derived alleles
- only homozygous loci
- only highly conserved sites
- only coding sites

What do we expect in the Apennine bear?



What do we expect in the Apennine bear?

HIGH impact mutations



1. Predict deleteriousness - Other tools/scores



CADD

*Combined Annotation-Dependent
Depletion*



PhyloP



SIFT

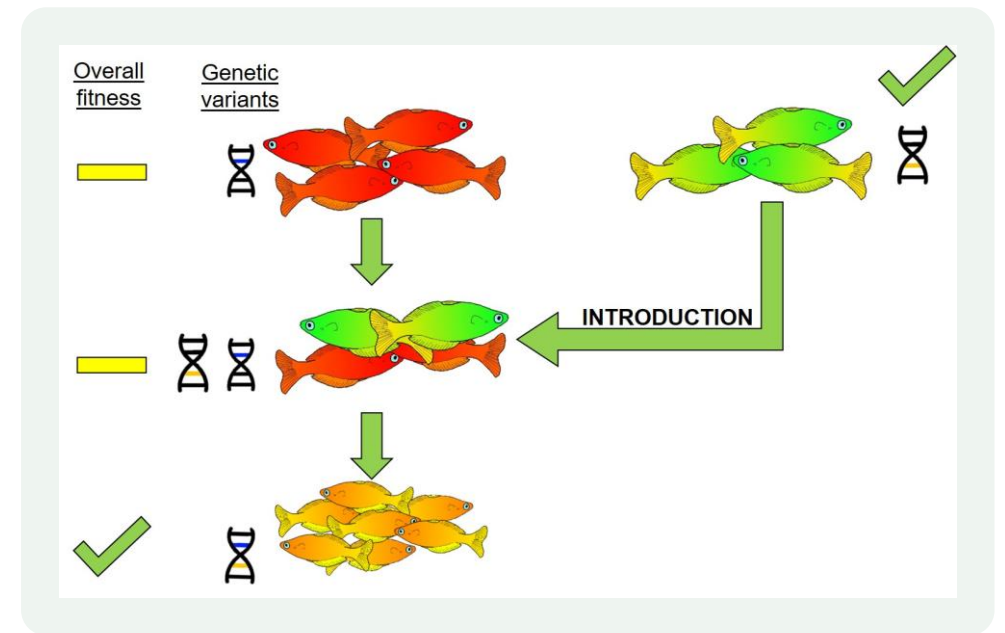
Sorting Intolerant From Tolerant

LiftOver

For ultraconserved elements, transfer information from model species to closely-related taxa

Genetic load and conservation

- Interestingly, low correlation between IUCN Red List status and genetic load^{1,2}
- Genetic load can be useful for:
 - guide captive breeding
 - genetic rescue
 - reintroduction programmes



¹ Díez-del-Molino et al. (2018) *Trends Ecol Evol*

² Hedrick, P. W. & Garcia-Dorado (2016) *Trends Ecol Evol*

Current challenges



The ongoing debate over large vs. small population for genetic rescue



Not all mutations are always deleterious



Dominance and gene interactions



Lack of experimental evidence

Recommended readings



- Bertorelle, G., Raffini, F., Bosse, M., Bortoluzzi, C., Iannucci, A., Trucchi, E., ... & Van Oosterhout, C. (2022). Genetic load: genomic estimates and applications in non-model animals. *Nature Reviews Genetics*, 23(8), 492-503. <https://doi.org/10.1038/s41576-022-00448-x>
- Dussex, N., Morales, H. E., Grossen, C., Dalén, L., & van Oosterhout, C. (2023). Purging and accumulation of genetic load in conservation. *Trends in ecology & evolution*, 38(10), 961-969. <https://doi.org/10.1016/j.tree.2023.05.008>